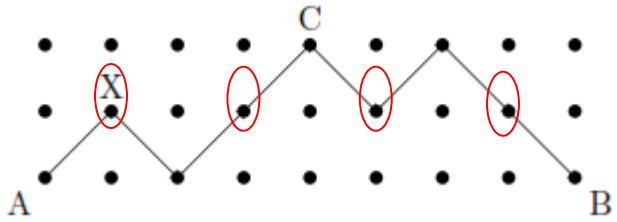
# Exercise on Paths 2

- Consider the following picture. You are only allowed to move ↗ (one to the right and up) and ↘ (one to the right and down) only. In how many ways you can get from A to B such that:




- **(i)** There are no restrictions. **Hint**: Note that in order to get from A to B you have to pass from some certain points like X.
- The path has to pass from the middle point (e.g., X) of every other column. There are 4 such points (in red). The path between any two successive red points has only two options (top or bottom row). Total number of paths $2^3$.
- **(ii)** You want to pass from C.
- 2 (ways from A to C) x 2 (ways from C to B) = 4